

CURRICULUM VITAE

Taekyung Kang

Education

Yale University, New Haven, CT	08/2025 – present
Ph.D. in Biomedical Engineering Advisor: Professor Evelyn Lake	
University of California – Irvine, Irvine, CA	09/2018 – 06/2024
B.S. in Biological Science B.S. in Bioinformatics	

Research and Academic Experience

01/2025-present	Graduate Researcher, Biomedical Engineering, Yale University, New Haven, CT Department of Radiology & Biomedical Imaging Advisor: Prof. Evelyn Lake
08/2024-12/2024	Graduate Researcher, Biomedical Engineering, Yale University, New Haven, CT Department of Internal Medicine Advisor: Prof. Fadi Akar
06/2023-09/2023	Research Intern, Cardiovascular Institute, Stanford University, Stanford, CA Department of Cardiovascular Institute Advisor: Prof. Ronglih Liao
09/2021-07/2024	Undergraduate Researcher, University of California – Irvine, Irvine, CA Biological Chemistry, School of Medicine Advisor: Prof. Cholsoon Jang
05/2021-09/2021	Research Intern, Molecular Biology, Republic of Korea School of Medicine, Seoul National University Advisor: Yong-Tae Kwon

Teaching and Mentorship

09/2025-02/2026	Mentor at Program to Advance Training in Health & Sciences (PATHS) Mentoring undergraduates and graduates preparing for MD, MD/PhD, or PhD careers Yale University, School of Medicine
08/2025-12/2025	Teaching Fellow to Prof. Douglas Rothman Biomedical Engineering 5410: Physical and Chemical Basis of Bioimaging and Biosensing Yale University, School of Engineering & Applied Science
09/2023-06/2024	Teaching Assistant for Prof. Susan King Chemistry 51C: Organic Chemistry University of California – Irvine, Department of Chemistry
01/2021-03/2024	Undergraduate Student Mentor Korean Health Association (KHA) University of California – Irvine, Department of Chemistry

Professional Honors, Recognition, and Development

2026-2027	Computational Molecular & Functional Imaging Training (CMFIT) Program, Yale University
2026	Neurophotonics Summer Professional Development, Boston University
2025-Present	Advanced Graduate Leadership Program (AGLP) Fellow, Yale University
2025-Present	Physics, Engineering, and Biology (PEB) Fellow, Yale University
2024	Cum Laude Scholar, University of California – Irvine
2024	Academic Research Award, University of California – Irvine
2023	Excellence in Research Fellowship, University of California – Irvine
2022	University Research Opportunity Program (UROP) Fellow, University of California – Irvine
2022-2024	Dean's Honors List, University of California – Irvine
2018	Presidential Volunteer Service Awards

Academic Activity

Publications

- Kang T.***, Mandino F, Allen N, Mano O, Shamble P, Greenwood J, Lake E. M. (2025). Functional MRI compatible simultaneous cortical wide-field fluorescent Ca²⁺ and intrinsic oxy/deoxy-hemoglobin imaging. In: Proceedings of the Cape Town - 2026 ISMRM-ISMRT Annual Meeting and Exhibition, Cape Town, South Africa. Program #461-06-002.
- Mandino, F.*, **Kang, T.**, Shen, X., Horien, C., Papademetris, X., Strittmatter, S. M., & Lake, E. M. (2025). Longitudinal, simultaneous wide-field fluorescent Ca²⁺ imaging and fMRI in awake mice. *bioRxiv*, 2025.10.27.684863. <https://doi.org/10.1101/2025.10.27.684863>
- Rudokas, M. W.*, McKay, M.*, Wu, X., Granger, J., Williams, Y., Bögner, M., **Kang, T.**, Seyfried, A., Toksoy, Z., Cacheux, M., Young, L. H., & Akar, F. G. (2025). Transient AMPK activation by nutrient stress of high fat diet preserves cardiac electrophysiological stability and protects against arrhythmias. *bioRxiv*, 2025.06.11.658631. <https://doi.org/10.1101/2025.06.11.658631>
- Bae, H.*, Jung, S.*, Le, J., Tamburini, I., Kim, J., Wang, E., Song, W. S., Choi, W., Jang, K. H., **Kang, T.**, Lopez, M. L., Ramirez, C., Mohanty, I., Kelly, M. E., Kim, J., Kim, R., Park, S. H., Baek, J., Mendez, B., Petrus, P., ... Jang, C. (2025). Cross-organ metabolite production and consumption in healthy and atherogenic conditions. *Cell*, 188(16), 4441–4455.e16. <https://doi.org/10.1016/j.cell.2025.05.001>
- Lopez, M. L.*, **Kang, T.***, Espeleta, A., Rubtsova, V. I., Baek, J., Songcuan, J., Moyer, E. M., Kim, J., Song, W. S., Jung, S., D'Sa, N., Anica, A., Tran, E., Chun, Y., Choi, W., Jang, K. H., Kelly, M. E., Tamburini, I. J., Alam, Y. H., Le, J., ... Jang, C. (2025). Intestinal catabolism of dietary fructose promotes obesity and insulin resistance via ileal lacteal remodeling. *Science Advances: preprint*, 2026.08.08.670963. <https://doi.org/10.1101/2025.08.18.670963>
- Bae, H.*, Kim, B. R.*, Jung, S.*, Le, J., van der Heide, D., Yu, W., Park, S. H., Hilkin, B. M., Gansemer, N. D., Powers, L. S., **Kang, T.**, Meyerholz, D. K., Schuster, V. L., Jang, C., & Welsh, M. J. (2024). Arteriovenous metabolomics in pigs reveals CFTR regulation of metabolism in multiple organs. *The Journal of Clinical Investigation*, 134(13), e174500. <https://doi.org/10.1172/JCI174500>
- Kang, T.***, Bae, H., Jang, C. (2023) Elucidating how fructose affects gut lacteal integrity via microbiota dysbiosis. *Journal of Undergraduate Research in the Biological Sciences*, 62(8).

Poster

- Kenneth Rainin Foundation (KRF) 2026 Innovation Summit, San Francisco, CA: Brain network signatures of repeated DSS-induced ulcerative colitis in mice, July, 2026
- Organization for Human Brain Mapping (OHBM) Annual Meeting and Exhibition, Bordeaux, France: fMRI Compatible Simultaneous Cortical Wide-Field Fluorescent Ca²⁺ and Intrinsic Hemodynamics Imaging, June, 2026
- International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting and Exhibition, Cape Town, South Africa: Functional MRI compatible simultaneous cortical wide-field fluorescent Ca²⁺ and intrinsic oxy/deoxy-hemoglobin imaging, May, 2026
- Yale Biomedical Engineering Symposium, West Haven, CT: Application of multimodal WF imaging with fMRI to image multi-fluorescence, Mar, 2026
- Yale Biomedical Imaging Institute Dean's Workshop, New Haven, CT: Simultaneous wide-field optical microscopy (WFOM) and functional magnetic resonance imaging (fMRI) 2.0, Oct, 2025
- Suh Kyungbae Foundation (SUHF) Annual Symposium, Seoul, Korea: The role of cellular cross talk via exosomes in cardiac light chain amyloidosis, Sep. 2023
- Stanford Summer Internship Symposium, Stanford, CA: The role of cellular cross talk via exosomes in cardiac light chain amyloidosis, Aug. 2023
- Excellence of Research, School of Medicine at University of California – Irvine, Irvine, CA: Elucidating How Fructose Affects Gut Lacteal Integrity via Microbiota Dysbiosis, Apr. 2023
- Undergraduate Research Opportunity Program (UROP) at University of California – Irvine, Irvine, CA: Elucidating the effect of Fructose Control over KHK-C Kinase, Jan. 2022

Volunteer and Community Engagement

2020	Inje University Paik Hospital Medical Assistant
2019	Busan Medical Center Medical Assistant

Extracurricular Activities

2026-present	President, Yale Korean Tennis Club (YKTC)
2026-present	President, Korean Graduate Student Association (KGSA) at Yale
2017-2026	Fellow, Korean-American Scientists and Engineers Association (KSEA)
2025-2026	Board Member, Yale Korean Tennis Club (YKTC)
2025-2026	Board Member, Korean Graduate Student Association (KGSA) at Yale
2023-2024	Board Member, HACK-UC
2022-2024	Board Member, LIKELION at UCI
2017-2018	Fellow, Columbia University Science Honors Program